SCHOOL OF TECHNOLOGY

FINAL YEAR PROJECT II - SYSTEMS IMPLEMENTATION TRACK

ROLLOUT STRATEGY DOCUMENT

SYSTEM IMPLEMENTATION PLAN

Deployment, Migration, Training, and Change Management Rollout for the MedPlus Pharmacy POS & Inventory Management System

PREPARED FOR:

MedPlus Board of Directors, Project Sponsors & Investors

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1. Executive Summary & Rollout Approach

The transition of MedPlus Pharmacy from a manual, paper-based operational model to the digital **MedPlus Point of Sale (POS) & Inventory Management System** represents a key business upgrade. This document provides a detailed technical and operational blueprint to guide the system's execution. By utilizing a phased implementation strategy, MedPlus ensures complete data integrity, high staff readiness, and minimal operational downtime during the transition.

Our approach emphasizes a **14-day parallel conversion window**, robust user training, and structured contingency protocols. This plan is formatted according to academic and professional standards, serving as a roadmap for development teams, system managers, and investors.

2. Implementation Schedule

The implementation schedule spans a structured **6-week lifecycle**, divided into five distinct operational phases. This ensures that software deployment, database migrations, and training are executed sequentially with clear gates between phases.

Week 1 - Phase 1: Infrastructure Deployment

Hardware Provisioning & Server Setup

Procuring local POS terminals, barcode scanners, and thermal receipt printers. Setting up the production environment hosting in a containerized Docker structure.

Week 2 - Phase 2: Database Preparation & Seeding

Data Schema Migration & Catalog Import

Migrating the core PostgreSQL schema. Seeding structural tables (role-based access controls) and parsing historical inventories into SQL database registers.

Weeks 3 & 4 - Phase 3: Staff Training & Sandbox Testing

Role-Based Training Modules

Conducting interactive workshops for pharmacists, inventory managers, and system administrators using a secure, isolated sandbox environment.

Week 5 - Phase 4: Parallel Conversion Trial

Dual-System Operational Integrity

Running the digital POS system concurrently with the manual ledger books to audit accuracy, verify M-Pesa API response times, and resolve teething bugs.

Week 6 - Phase 5: Final Cutover & Go-Live

Legacy Ledger Decommissioning

Deactivating manual ledger logging. Archiving physical records, shifting operations entirely to the MedPlus POS system, and transitioning to the software maintenance cycle.

Phase / Milestone	Core Tasks	Duration	Responsible Party
1. Infrastructure Setup	Server build, Docker install, POS hardware delivery.	5 Days	Systems Architect / IT Ops
2. Data Migration	Clean legacy records, run SQL migration scripts.	4 Days	Database Administrator
3. Staff Training	Workshop runs, simulated checkout trials.	7 Days	Implementation Lead
4. Parallel Runway	Execute double entries, audit inventory balance.	10 Days	Pharmacists & Managers
5. Full Cutover	Lock old logs, verify system, open main gate.	2 Days	Project Manager / Lead Dev

3. Hardware & Software Installation Plans

Hardware Installation Specifications

To support the 2030 dark-glassmorphic frontend visual framework and processing routines, the on-site pharmacy layout requires standardized hardware endpoints. Each Point of Sale checkout counter will be equipped with the following certified terminals:

- **Local POS Terminal:** Windows-based or Linux PC/Tablet featuring minimum 8GB RAM, Intel Core i3 CPU, and a 1080p display to render backdrop filters and animations smoothly.
- **Barcode Scanner:** USB omni-directional laser scanner compatible with standard 1D/2D barcodes (EAN-13 format).
- **Thermal Receipt Printer:** 80mm high-speed thermal receipt printer utilizing standard ESC/POS command sequences.
- **Networking Infrastructure:** Secure dual-band gigabit router configured with WPA3 encryption, ensuring isolated VLAN divisions for payment traffic and internal databases.

Software Installation & Orchestration

The MedPlus Pharmacy POS system is deployed using Docker containerization, which isolates the application stack and guarantees cross-platform consistency. The application architecture consists of three core containers:

1. **Web & API Container:** Houses the compiled Go backend binary. The server serves static frontend files from the `web` directory and runs backend handlers to resolve API requests.
2. **Database Container:** Runs a secured instance of PostgreSQL. Operational schemas, RBAC indexes, transaction logs, and customer matrices reside in persistent volumes mapped to host storage.
3. **Reverse Proxy Container (Optional / Production):** Nginx container managing SSL termination (HTTPS) and routing traffic to the Go API port.

> **Production Environment Settings**
>
> For deployment, the `.env` file must override all demo seeds. Ensure that `APP_ENV=production` is active, `SEED_DEMO_DATA=false` is set to prevent seeding mock data, and `JWT_SECRET` is updated to a cryptographically strong 256-bit string.

4. System Conversion Strategy

Transitioning the pharmacy from manual logbooks to the digital database requires a robust data migration strategy. A direct cutover ("cold turkey" approach) presents too high a risk for daily patient prescriptions. Instead, a **Phased Parallel Conversion Strategy** will be used:

Data Migration & Activities of Conversion

- **Inventory Extraction:** Legacy stock levels, cost thresholds, batch barcodes, and drug names are extracted from physical documents or Excel sheets and compiled into a sanitized CSV template.
- **Validation & Mapping:** Data formats are verified. Low dollar prices are mapped to realistic Kenyan Shillings (Ksh), and categories (OTC, Prescription, Cosmetics) are assigned to each item.
- **Database Ingestion:** A custom Python database wrapper connects to the PostgreSQL container to insert inventory lists, customer accounts, and supplier records, ensuring database consistency.
- **Verification Check:** Running SQL count queries against original CSV lists to verify zero record losses during ingestion.

Parallel Run & Gate Reviews

Beginning in Week 5, the pharmacy staff will operate under a dual-entry scheme. When a customer checkout occurs:

1. The pharmacist processes the sale on the physical paper logbook.
2. The pharmacist simultaneously enters the transaction on the digital MedPlus POS interface, generating the M-Pesa STK push callback or recording cash metrics.

At the end of each shift, managers will cross-verify total revenue and stock balances between the paper ledger and the digital reports screen. Discrepancies will be flagged to developers. Once the digital system completes 7 consecutive days of 100% database accuracy, the manual ledgers will be decommissioned.

5. Training Plan

A software system is only as effective as the staff operating it. The training plan is designed to build system confidence and operational efficiency across all pharmacy roles. Training will be divided into three methodologies:

Training Methodologies

- **Simulated Workstations (Sandbox Trials):** A duplicate staging environment is seeded with sample medicines. Staff can scan real items and complete simulated checkouts without affecting production data.
- **Role-Specific Classrooms:** Tailored training sessions designed around specific role permissions. This keeps pharmacists focused on POS, customers, and prescriptions, and admins focused on audits and RBAC.
- **Supervised Shadowing:** During the first 3 days of parallel operations, a tech supervisor will shadow checkout counters to help resolve real-time staff queries.

Training Curriculums by Role

Staff Role	Core Training Modules	Staging Assessment Requirement
Pharmacists & Cashiers	Cart checkout, barcode scanning, Rx validation, M-Pesa push triggers, receipt printing.	Process 10 simulated sales (including 3 Rx validations and 3 M-Pesa transactions) under 5 mins.
Stock Managers	Product directory CRUD, batch registry, restocking PO status transitions, supplier mapping.	Create a purchase order, receive items, and verify stock level updates in inventory.
System Admins	User creation, dynamic RBAC permission overrides, database overview, password resets.	Revoke access to "Reports" for a test user and unlock the database schema using admin PIN code.

6. Software Maintenance Plan

Post-deployment operations are governed by a proactive maintenance schedule to ensure high database integrity and backend performance.

- **Database Backups:** Automated daily backups are scheduled at midnight local time using a cron daemon. PostgreSQL database dumps are compressed and written to secure, encrypted volumes.
- **Dependency Patching:** Go package dependencies and Docker container base images are audited monthly for security vulnerabilities and updated to stable releases.
- **API Performance Audits:** Backend handler response times for critical checkout endpoints (like `/api/sales` and M-Pesa callback webhooks) are monitored. High-latency queries are optimized with indexing.
- **Security Integrity Checks:** JWT signature keys are scheduled for rotation every 90 days. Rate limiting blocks on authentication points are audited.

7. Change Management Plan

Digital transformation often creates friction. Our change management strategy focuses on minimizing disruption, addressing staff concerns, and ensuring smooth adoption of the MedPlus system.

Addressing Staff Transition & Resistance

- **Emphasizing Efficiency:** Training will demonstrate that the POS system handles inventory decrements and purchase ordering automatically, saving staff hours of manual calculations and logbook recording.
- **Simplifying Tasks:** The 2030 glassmorphic UI includes automated visual alerts (like low-stock flags and expiring medicine warnings) to simplify daily work.

Contingency & Fallback Procedures

In the event of network disruption, the pharmacy cannot stop operations. The following protocols will apply:

1. **Internet Outage:** If external internet drops, the local router network remains active. Staff can access the local server container via Wi-Fi to process cash sales. M-Pesa STK pushes will be suspended, and cash/card payments will be used.
2. **Power Failure:** POS terminals are connected to Uninterruptible Power Supplies (UPS) that provide 30 minutes of backup runtime, allowing transaction queues to clear. A backup generator supports the local network router.
3. **Critical Server Crash:** In the rare event of a container failure, a secondary local backup server can be brought online within 15 minutes, restoring the database from the last night's backup.